

Stage 3: Model Building

C22 Retail Fade Strategy

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*Route taken: **report the negative result**. All figures are reverseProfit per lot; the \$7.00/lot cost is already inside the target and is applied exactly once (asserted in code, max deviation 4.55×10^{-13} over $n = 7,277$). Hurdle is exactly zero. Common split: fit and select on C33–C52 ($n = 694$), evaluate on C53–C66 ($n = 6,583$; $n = 6,582$ after the C66 exclusion described in the data note). Our own split is reported alongside. All intervals are 95% bootstrap clustered by `accountId` unless explicitly labelled otherwise; earlier drafts clustered by `traderKey` and are restated in Section 10.*

1. Result

There is no exploitable edge at this cost level that we can demonstrate. We can state how large an edge we would have detected, and — added since the first draft of this submission — we can state that most of what looked like signal was leakage.

- **Eleven of our twenty model features were future-looking**, through a mechanism the cohort-wide `slPrice` guidance does not cover. The submitted model had that information and still lost money.
- **Two of the three effects clearing the corrected detection floor are contaminated** — one is among those eleven, the other a screened feature we had already dropped. One clean effect remains, and it is not behavioural (Section 7).
- No admissible feature survives Benjamini–Hochberg correction. The corrected trial denominator is $m = 4,626$, not the 101 endpoint entries originally logged; **0 of 90 scalar outcome tests survive**.
- **Zero of seven** coverage levels, from 1% to 85%, produce an observed effect exceeding their own minimum detectable effect.
- The mandated training split cannot calibrate an executable threshold: maximum training support at any executable threshold is $n = 4$, against our pre-registered floor of $n \geq 30$.
- The frozen model shows +10.433/lot equal-weighted but –33.849/lot size-weighted, losing \$3,556.81 against DO NOTHING’s \$0. It also fails to beat FADE EVERYTHING: intervals overlap almost entirely.

Sections 2–9 evidence each claim. Section 10 is a register of every figure restated since the first draft.

2. Leakage: a general mechanism, not a field

C22 ruled `slPrice` inadmissible because it is written at close. We removed the stop-derived features accordingly. Re-auditing under the Stage 3 `tpPrice` guidance, we found that the ruling identifies a *field* where the underlying defect is a *mechanism*.

Positions overlap. Measuring directly: **63.34% of five-trade lookback windows (1,529 of 2,414 computed) contain at least one prior trade that was still open at the current trade’s entry**. For those windows, every close-time property of the prior trade — realised profit, close price, duration, exit type, and the stop — was undetermined at the moment the current trade opened. Any rolling feature over prior trades that reads such a field is future-looking, whether or not the field is `slPrice`.

The standard leakage test does not detect this. C22 lists recompute-after-deleting-later-rows as acceptable, and two Stage 2 submissions used it. Neither simple form works under concurrent positions:

- Filtering prior rows by *entry* timestamp leaves a prior trade that opened earlier and closed later inside the window, with its close-time fields intact. On our own set this returned **0 mismatches across all 11 contaminated features** — a clean pass on a contaminated model.
- Filtering by *close* timestamp over-rejects: dropping a still-open prior changes window composition, so admissible open-time features mismatch too.

The construction that works is **field-class-aware**: apply the entry filter to open-time fields (*openPrice*, *amount*, *direction*, *challenge_type*, *open timestamp*) and the close filter to close-time fields (*profit*, *closePrice*, *slPrice*, *tpPrice*, *duration*, *exit type*, *realised return*). Under that test, **all 9 clean features are bit-identical and 10 of 11 contaminated features mismatch**; the eleventh, *prior_campaigns_x_loss_streak_ge_2*, is sample-degenerate (62 of 64 zero in the causal sample) rather than admissible. This test now runs in the production build path, not only in the suite.

Contaminated inputs (11 of 20): *loss_streak*, *win_streak*, *pnl_ewm*, *size_after_loss_delta*, *sl_usage_rate_5*, *manual_exit_rate_5*, *pnl_pct*, *dd_from_peak_pct*, *log_dt_close*, *prior_campaigns_x_loss_streak_ge_2*, and *same_direction_reentry*. Only two involve stop or take-profit fields; the other nine were invisible to a field-based audit.

What it does to the result. We refit on the 9 clean features under an identical selection procedure (α unchanged at 3162.28). Coverage collapses from 2.69% to **0.076% — five trades**, below our own pre-registered $n \geq 30$ floor. The contaminated features were carrying the decision function; without them there is almost nothing to act on. We therefore issue no forecast for the clean refit (Section 8) and freeze the disclosed model instead.

The conclusion is unchanged and better supported. A model with access to future information lost \$3,556.81 against doing nothing. The negative result was never load-bearing on the contamination.

3. Model, and why

Ridge on reverseProfit per lot, target winsorised at the 1st/99th percentile computed on training folds only. Inner expanding-campaign CV selected $\alpha = 1000$; we froze $\alpha = 3162.28$, one half-decade more regularised, because Stage 2 documented sharp cross-fold instability and a composite of weak signals should be shrunk harder than CV alone suggests.

Feature set: V2. Twenty raw features, twenty-one transformed columns. The figure of 24 in our first draft was the screened admissible *family*; the frozen `predict()` path retains 20, with the four cross-campaign history fields deliberately omitted. Artifact coefficients and the prediction matrix match by name and position; this was verified, not assumed. *trader_prior_sl_discipline*, listed as retained in the first draft, is **not present in V2** at all.

V1 retained the four history features with an explicit cold-start indicator; V2 dropped them. V1 was rejected on evidence:

- Under 79.39% cold-start in the evaluation window, *is_cold_start* carries a ridge coefficient of 2.705 against 0.012–0.547 for the four history features. V1 is fitting the cold-start partition, not trader history.
- Full-evaluation intervals overlap: V1 $[-49.44, 43.16]$, V2 $[-58.43, 37.92]$ (*traderKey*-clustered, retained as performed at the time of the comparison).
- On the 20.57% of rows that carry trader-specific history — the only place history can help — V1 loses to V2 (-2.579 , $n = 182$ vs $+10.031$, $n = 155$).

V2 was preferred on set-level parsimony, with no history-specific benefit demonstrated. We note in hindsight that one of the dropped fields, *trader_prior_survival*, is contaminated (it consumes *last_close* minus *first_open*), so the drop was correct for a reason we did not have at the time.

The 21st column. *challenge_type* enters as a two-level one-hot, {11, unknown}, where *unknown* absorbs 61% of rows and conflates a recorded category with an absent field. It is admissible — known at open — but it is not a behaviour, and we do not defend it as one. Removing

it moves the headline from +10.433 to −0.874/lot, with overlapping intervals ($[-46.12, 66.83]$ and $[-53.45, 48.84]$), both containing zero. The mechanism is entirely selection: 13 trades change at the abstain boundary and the 12 added average roughly −167/lot. This is the same defect as Section 5 — no executable threshold has more than $n = 4$ support. Resolving the `challenge_type` unknown remains a recommendation to C22.

Inadmissible features. `sl_distance_pct`, `sl_widening_delta`, `has_sl` and `has_tp` are excluded per C22’s ruling. We attempted the recovery C22 invited — reconstructing the opening-ticket stop from the raw fills — and **disproved it**: across 4,543 multi-fill primary-era positions, `slPrice` changes between first and last fill in **73.12%** of cases (`tpPrice`, 79.84%); the collapse takes the first *non-null* value, so where a stop was added mid-trade that value is itself mid-trade; direction-violation rates are 11.98% on the first fill against 16.51% on the last, both far above the near-zero rate a genuine open-time field would show; and violating rows are systematically profitable (−68.53/lot, 57.12% win rate), meaning the violation flag alone encodes outcome. The opening stop is not recoverable from this data.

Decision rule, as executed by `predict()`:

$$\text{FADE} = (\text{score} > \max(\theta_{\text{sel}}, \text{hurdle})) \wedge (\text{win_streak} \leq 1) \wedge (\text{trades_per_hour} \leq 60)$$

with $\theta_{\text{sel}} = -10$ and `hurdle` = 0. Both abstain overrides derive from Stage 1–2 findings. Their first Git-recorded commit is 06efdd59, dated 2026-08-21 — the submission date — so version control does not by itself establish pre-registration, and we disclose that rather than claim it.

4. Backtest on the common split

	n acted	coverage	equal-wtd	size-wtd	total \$
Model (overrides)	177	2.689%	+10.433	−33.849	−3,556.81
Model (no overrides)	183	2.780%	+10.452	—	−3,534.65
DO NOTHING	0	0.000%	0.000	0.000	0.00
FADE EVERYTHING	6,582	100.00%	+5.434	−3.001	−12,108.71

Model `accountId` CI: equal-weighted $[-46.117, 66.826]$, size-weighted $[-70.260, 2.964]$. FADE EVERYTHING equal-weighted CI $[-8.572, 20.466]$. Coverage is 177/6,582 under the C66-excluded convention adopted throughout; earlier drafts mixed a C66-excluded numerator with a C66-inclusive denominator (Section 10). Acted lots total 105.08; $-3,556.81/105.08 = -33.8486$, and both figures use the same C66-excluded base.

The model beats neither baseline. Against DO NOTHING it loses \$3,556.81. Against FADE EVERYTHING it cannot be separated: +10.433 $[-46.12, 66.83]$ on 177 trades against +5.434 $[-8.57, 20.47]$ on 6,582, intervals overlapping with the baseline’s roughly four times tighter. C22 requires beating both; we beat neither, and report that as the finding rather than an omission.

The two metrics disagree, and this is a property of the data. Both policies show it: the model at +10.433/lot but −\$3,556.81, and fade-everything at +5.434/lot but −\$12,108.71. Small and large trades carry opposite information, so any size-blind policy produces a positive per-lot number and loses money. The model captures 29.37% of the total fade-everything dollar loss while acting on 2.69% of positions and 2.60% of volume — $10.9\times$ its position share. It is selecting large-lot losers. We report the equal-weighted figure because it is the requested metric, and the size-weighted figure because it is what happens to the book.

Per-campaign. The result is not distributed. C53 gives +173.557 on $n = 5$, the only positive campaign whose CI excludes zero; removing it halves the result to +5.691 ($n = 172$). C59 is significantly *negative*: −99.990 on $n = 21$. Seven of the thirteen non-excluded campaigns are negative. Leave-one-campaign-out over all fourteen, C66 included, moves the headline equal-weighted result between −1.355 (dropping C64) and +25.081 (dropping C59); the size-weighted result stays negative under every drop.

Our own split. Running the frozen model unchanged over the Stage 2 walk-forward design — no refit, evaluation window only — Track A acts on 6 of 2,190 (0.274%) for +83.005/lot, CI $[-117.77, 289.44]$, −\$48.87; Track B acts on 134 of 4,545 (2.948%) for −7.573/lot, CI $[-70.38, 61.38]$,

−\$4,724.18, size-weighted −55.409 with CI [−94.24, −15.78] excluding zero. Neither track beats DO NOTHING's \$0. These figures supersede the fold-refit numbers in our first draft, which were not a no-refit evaluation of the frozen artefact.

In-sample to out-of-sample gap. Training +50.897 on $n = 4$, CI [−37.213, 142.123]; evaluation +10.433 on $n = 177$. The gap is −40.571/lot. We report it because C22 requires it and because the gap size predicts Stage 4 survival, but four observations do not anchor anything — which is the subject of Section 5. The in-sample point estimate does not clear its own corrected detection floor either.

5. The mandated split cannot calibrate a threshold

A structural property of the common split, not a tuning failure.

- **The training window is 694 positions across 330 accounts** against 6,582 evaluation positions (6,583 before the C66 exclusion) — roughly one-tenth the test size. That is 34.70 positions, or 16.50 accounts, per admissible feature at the corrected count of 20.
- **79.39% of evaluation positions belong to traders with no history in the training window.** Four history features are structurally uncomputable for four in five evaluation rows.
- **Corrected training MDE is \$53.02–\$61.03/lot** under `accountId` clustering (winsorised and raw respectively). **Three of 48** admissible pooled effects observed anywhere in Stage 2 exceed \$53.02 — so this split could not have detected 45 of the 48 effects we observed. Two of those three are contaminated (Section 7). Fitting across the C53 regime break costs a further \$12.27/lot versus training within the post-break era.
- **No executable threshold has adequate support.** Selection maximised training mean subject to $n \geq 30$ and chose $\theta = -10$ ($n = 88$). But the economic hurdle forbids fading a negative predicted return, so every negative threshold is overridden and the true operating point is `score > 0`, where training support is $n = 4$. Re-running selection over the executable grid [0, 500] confirms $n = 4$ is the maximum available at any executable threshold. **The pre-registered $n \geq 30$ floor is unsatisfiable on this split.** Execution reduces to `score > 0` plus the two abstain overrides.

6. Power: what we would have detected

Target	n acted	achieved	observed rP/lot	<code>accountId</code> CI	MDE	clears?
1%	66	1.003%	+2.279	[−98.88, 104.67]	191.50	no
2.7%	177	2.689%	+10.433	[−46.12, 66.83]	116.94	no
5%	329	4.999%	+23.284	[−26.72, 77.10]	86.83	no
10%	658	9.997%	+22.124	[−20.76, 65.53]	66.66	no
25%	1,646	25.008%	+32.118	[2.65, 61.99]	38.37	no
50%	3,291	50.000%	+21.675	[1.43, 42.57]	27.41	no
100%	5,615	85.308%	+13.139	[−1.79, 28.42]	23.13	no

Coverage levels sweep the execution hurdle while retaining both abstain overrides; the frozen model sits at the 2.689% row, and coverage caps at 85.31% because the overrides are unconditional. $MDE = 2.801585 \sigma \sqrt{DEFF/n}$ at $\alpha = 0.05$, power 0.80, with σ held at primary-era dispersion (raw 555.33; winsorised 482.45) so the calculation does not reuse the selected subset's realised variance. Acting-subset ICC estimates are negative at mean cluster size ≈ 2.7 ; the resulting $DEFF < 1$ is a small-sample artefact and anti-conservative, and the training-window ICC is positive, so the floor at zero is retained.

Zero of seven coverage levels clear their own detection floor. The closest is 25% coverage: 32.118 against 38.37. Narrowing coverage to find a state where edge holds raises the detection floor as $1/\sqrt{n}$, faster than narrowing plausibly raises the effect.

Two criteria disagree at high coverage, and we state why. At 25% and 50% coverage the realised interval excludes zero while the effect still fails its MDE. These measure different things — MDE is prospective power at a pre-specified effect size, the interval is realised precision — and the

resolution is the size asymmetry of Section 4: at high coverage the per-lot estimate is significantly positive while the policy loses money in dollars. Neither criterion supports deployment.

The score does not rank trades. If it ordered by fadeability, mean edge would decline monotonically as lower-scored trades are added. Instead the top 1% of scores returns +2.279/lot on the widest interval in the table, while mean edge peaks at 25% coverage. The evaluation score distribution (median -29.92, p75 -15.29, p99 +27.53, s.d. 30.36) shows the model predicting a loss on three-quarters of the window.

7. Trials log and multiple-testing correction

Our first draft logged **101** trials. Auditing the true search space — feature sets, model variants including every α in the ridge grid, thresholds, and abstain variants — gives **4,626 pre-freeze attempts**, a denominator 45 $\times$ larger. We report the larger figure because it is the correct one and because it strengthens the null. Under Benjamini–Hochberg at $q = 0.05$ with $m = 4,626$: **0 of 90 scalar outcome tests survive**. The only survivors under the original narrow denominator were tests of the now-inadmissible `sl_distance_pct` — **the leak was producing our only statistically defensible results**, a pattern Section 2 shows was general rather than specific to that field.

Contamination explains the apparent signal. Of the three effects clearing the corrected \$53.02 floor, two are contaminated: `manual_exit_rate_5` (64.912/lot, $n = 121$), which is among the eleven, and `trader_prior_survival` (70.630/lot, $n = 373$), a screened feature absent from V2 and whose `accountId` interval $[-1.069, 147.010]$ now includes zero in any case. One survives as clean.

The one clean detectable effect is not behavioural. `ip_cluster_size` — 67.775/lot, $n = 369$, CI [3.919, 144.992] — consumes only open-time context metadata and is present in V2. It does not survive correction at $m = 4,626$. More importantly, it is a statement about *account identity*, not trader psychology: accounts sharing an IP cluster behave differently. We do not offer it as a fading edge. We offer it to C22 as a multi-accounting and abuse-detection signal, alongside the Stage 2 Appendix E recommendations.

Post-hoc compliance sensitivity, segregated from the pre-freeze denominator and logged as such: the drop-2-contaminated ablation, the drop-challenge_type ablation, the fixed-acted-set counterfactual, and the clean-9 refit. We also record `prior_campaigns_x_loss_streak_ge_2` as sample-degenerate — it consumed a trial and contributes nothing.

Six conditional states were pre-registered before running and tested under two threshold provenances: S1, thresholds selected on training campaigns ≤ 52 only; and S2, thresholds carried forward from Stage 1–2, derived on C53–C65 and therefore leaky under the mandated split. No state survives correction under either. S2 states 3 and 4 clear zero unadjusted — reported as the negative control that demonstrates why the trials log exists, not as a finding.

8. What we freeze, and what we forecast

Frozen artefact: `artifacts/stage3_v2.json`, SHA-256 7d77eb8af7cde7bdef856f0d5a47d28a5881795da0a4cde4ed8421623bc359b1. V2 features, $\alpha = 3162.28$, hurdle 0, both abstain overrides. Entry point `predict(trades_df)`; deterministic, dependencies pinned with hashes, one documented command. The suite includes a banned-column assertion in the production build path, the field-class-aware causal rebuild of Section 2, a strict ordered-column contract against a persisted 21-column manifest, and row- and column-permutation invariance tests. Two clean-clone reproductions, both byte-identical on the artefact: the submission tag `stage3-submission-2026-08-21` at 119.46s with 20 tests, and the compliance tag `stage3-compliance-2026-08-28` (commit 80a6f19) at 102.99s with 28 tests, the eight additions being the compliance tests just listed. a strict ordered-column contract against a persisted 21-column manifest, Clean-clone reproduction from the frozen tag: 119.46s, 20 tests, artefact byte-identical; current worktree 98.37s, 28 tests.

We freeze it **as-is and disclosed as uncalibrated**: the operating point has $n = 4$ training support, 11 of its 20 inputs are contaminated, and it loses money. An always-abstain model would tie DO NOTHING rather than beat it, and would generate nothing for Stage 4 to score. We do not

recommend deploying it under any circumstances.

Not frozen. The clean-9 refit exists on a separate branch as disclosed remediation, artefact `stage3_v3_clean.json`. We do not submit it: at $n = 5$ acted trades it falls below our own pre-registered support floor and has no interpretable performance. A size cap in the decision rule would improve the dollar outcome, and we have not adopted it — that rule was chosen after observing the size-weighted result, which is precisely the behaviour the trials log exists to catch.

Stage 4 forecast. Registered against the frozen artefact after evaluation results were viewed. Supersedes the interval in our first draft.

Equal-weighted mean reverseProfit per lot on unseen campaigns: **0.00**, 95% interval $[-56.5, +56.5]$. Coverage: **under 5%**, threshold-driven and unstable across campaign blocks (0.27%–2.95% across our own tracks). Size-weighted mean: **negative**, point estimate $-33.85/\text{lot}$. We expect the model **not** to beat DO NOTHING in absolute dollars.

No forecast is issued for the clean-9 refit. At $n = 5$ it falls below our pre-registered $n \geq 30$ floor and we decline to register a number we cannot defend.

Derivation. The point estimate is zero because nothing in the analysis supports another value: no feature survives correction at $m = 4,626$, and no coverage level produces an effect above its own MDE. The interval is the half-width of the acting-subset `accountId`-clustered CI $[-46.117, +66.826]$, half-width 56.47, re-centred on zero. Our first draft gave $[-62, +62]$ from a `traderKey`-clustered interval; that figure is superseded. We forecast a null rather than the observed $+10.43$ precisely because our own analysis shows $+10.43$ is indistinguishable from zero.

9. Where it fails

1. **Leakage.** Eleven of twenty inputs are future-looking. Removing them collapses coverage to five trades.
2. **Size selection.** Equal-weighted positive, size-weighted -33.849 . The model concentrates into large-lot losers — the Stage 1 sizing paradox reappearing at the decision layer.
3. **Calibration.** The operating point rests on four training observations.
4. **Concentration.** Half the equal-weighted result comes from one five-trade campaign.
5. **Artefact dependence.** The headline's sign is set by a two-level dummy on a 61%-unknown metadata field, via 13 trades at the abstain boundary.
6. **No ranking power.** The top 1% of scores returns $+2.279/\text{lot}$, below every broader operating point.
7. **Regime dependence.** Training across the C53 break costs $\$12.27/\text{lot}$, and Stage 4 campaigns lie further still from the training window.
8. **Baselines.** Beats neither DO NOTHING nor FADE EVERYTHING.

10. Corrections register

Every figure restated since the first draft, with the reason. Numbers appearing only in the superseded PDF are enumerated in `report/number_crosscheck.md`. The superseded PDF itself is retained unedited as a frozen binary.

Item	First draft	Corrected	Reason
Feature count	24	20 raw / 21 columns	24 was the screened family; V2 retains 20
Contaminated inputs	2 (stop fields)	11 of 20	Field-class-aware rebuild (Section 2)
trader_prior_sl_*	retained	absent from V2	Not in the frozen feature list
Trial denominator	101	4,626	Endpoints counted, not attempts
Training MDE	\$65–78/lot	\$53.02–\$61.03/lot	accountId clustering
Detectable effects	1 of 48	3 of 48 (>\$53.02)	Corrected floor; 2 of 3 contaminated
Clustering unit	traderKey	accountId	C22 mandates account-level
Model EW CI	[−52.786, 72.331]	[−46.117, 66.826]	Clustering unit
Model SW CI	[−74.360, 1.166]	[−70.260, 2.964]	Clustering unit
Size-weighted /lot	−13.353 in places	−33.849	Earlier figure mixed a C66-inclusive lot base with C66-excluded dollars
Coverage	178/6,583 = 2.704%	177/6,582 = 2.689%	C66 now excluded from numerator and denominator alike
Own-split figures	fold-refit	no-refit on frozen model	First draft was not an evaluation of the frozen artefact
Test count	19	20 frozen / 28 worktree	Leakage, contract and permutation tests added
Stage 4 interval	[−62, +62]	[−56.5, +56.5]	Re-derived on accountId CI
Commission form	+0.0020 + 34.9917a	sign convention noted	Raw field is signed opposite; positive-cost convention used throughout

Data note. C66 collapses 1,911 fills into a single position whose amount is the sum across fills (462.30 lots, against a maximum collapsed amount of 6.50 lots in C53–C65), contributing −\$4,019.40. Sum aggregation remains correct for ordinary partial-close fills and no other primary-era multi-fill position is affected; C66 is a corrupted test export and is now excluded identically from numerators, denominators, dollar totals and lot bases across all three policies. Campaigns C1–C32 contain no trades in this dataset — the collapsed set spans C33–C66, $n = 7,277$. Cost application was asserted exactly once in code (max residual 4.55×10^{-13} over $n = 7,277$), so no Stage 2 restatement is required. Commission is deterministic at $0.0020 + 34.9917 \times \text{amount}$, $R^2 = 0.99984$, $n = 6,582$.